The arithmetic root of degree n

Undoing a power: what the n-th root means, when it exists, and the four properties that let you work with it.

LESSON 28–29 | 2 × 40 MIN | ALGEBRA 8, §8 | STAGE 9 · 1.1

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Define the arithmetic root of degree n and say when it exists.
- Use the four properties of roots to simplify.
- Take a factor out of a root and put one back in.
- Rationalise a simple denominator.

TEXTBOOK

National	§8, pp. 39–41
Cambridge	Learner's Book pp. 10–14
Workbook	Workbook 1.1

01 Explanation

The definition

DEFINITION

For $a \geq 0$ and a natural number $n \geq 2$, the **arithmetic root of degree n** of a is the **non-negative** number whose n-th power is a:

$$\sqrt[n]{a} = b \text{ means } b \geq 0 \text{ and } b^n = a$$

So $\sqrt{16} = 4$ — and only 4. The number -4 also squares to 16, but the *arithmetic* root is the non-negative one by definition.

WHEN DOES IT EXIST?

For **even** n the radicand must satisfy $a \geq 0$ — you cannot take $\sqrt{-9}$. For **odd** n any a is allowed: $\sqrt[3]{-8} = -2$.

The four properties

For $a \geq 0, b \geq 0$:

$$\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b} \qquad \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}, \; b > 0$$

$$(\sqrt[n]{a})^k = \sqrt[n]{a^k} \qquad \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

Read the first one right to left and you can multiply roots together; read it left to right and you can **take a factor out**:

$$\sqrt{50} = \sqrt{25 \cdot 2} = \sqrt{25} \cdot \sqrt{2} = 5\sqrt{2}$$

THERE IS NO RULE FOR A SUM

$\sqrt{a + b}$ is **not** $\sqrt{a} + \sqrt{b}$. Check it: $\sqrt{9 + 16} = 5$, but $\sqrt{9} + \sqrt{16} = 7$.

Rationalising the denominator

A root in the denominator is untidy. Multiply top and bottom by the same root:

$$\frac{1}{\sqrt{3}} = \frac{1 \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}} = \frac{\sqrt{3}}{3}$$

This is the fundamental property of a fraction again — the same tool as in Quarter I.

$\sqrt{a} \rightarrow a^{(1/2)}$

$\sqrt[3]{a} \rightarrow a^{(1/3)}$

$\sqrt[4]{a} \rightarrow a^{(1/4)}$

$\sqrt[n]{a^m} \rightarrow a^{(m/n)}$

a root is a power — the same object, two notations

Every root can be written as a power, which is the bridge to the next lesson.

02 Worked examples

EXAMPLE 1 ***Simplify*** $\sqrt{72}$

1. $72 = 36 \cdot 2$

   Look for the largest square factor.

2. $\sqrt{72} = \sqrt{36} \cdot \sqrt{2}$

   Property 1.

3. $= 6\sqrt{2}$

ANSWER

$6\sqrt{2}$

EXAMPLE 2 ***Simplify*** $\sqrt{12} \cdot \sqrt{3}$

1. $\sqrt{12} \cdot \sqrt{3} = \sqrt{36}$

   Property 1, read right to left.

2. $= 6$

   A whole number — worth checking for every time.

ANSWER

6

EXAMPLE 3 ***Write $\frac{6}{\sqrt{2}}$ without a root in the denominator***

1. $\frac{6 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$

   Multiply top and bottom by $\sqrt{2}$.

2. $= \frac{6\sqrt{2}}{2}$

   $\sqrt{2} \cdot \sqrt{2} = 2$

3. $= 3\sqrt{2}$

   Cancel the 2.

ANSWER

$3\sqrt{2}$

03 Interactive model

Move the index n and watch the same number appear as a root and as a power.

A root is a power

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64** | index n **3** | power m **1** | check: value^3 **64** | a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Root of degree n	n-darajali ildiz	Корень n-й степени
Arithmetic root	Arifmetik ildiz	Арифметический корень
Radicand (under the root)	Ildiz ostidagi ifoda	Подкоренное выражение
Index of the root	Ildiz ko‘rsatkichi	Показатель корня
Square root	Kvadrat ildiz	Квадратный корень
Cube root	Kub ildiz	Кубический корень
Irrational number	Irratsional son	Иррациональное число
Rationalise the denominator	Maxrajni irratsionallikdan qutqarish	Освободиться от иррациональности в знаменателе
Take a factor out	Ko‘paytuvchini ildiz oldiga chiqarish	Вынести множитель из-под корня
Simplify	Soddalashtirish	Упростить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sqrt{49}$ equals:

- 7
- ± 7
- -7
- 24.5

$\sqrt[3]{-27}$ equals:

- it does not exist
- -3
- 3
- -9

$\sqrt{8} \cdot \sqrt{2}$ equals:

- 4
- $\sqrt{10}$
- 16
- $2\sqrt{2}$

$\sqrt{9 + 16}$ equals:

- 7
- 5
- 25
- $3 + 4$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find* $\sqrt{36}$

   ANSWER 6

2. *Find* $\sqrt{81}$

   ANSWER 9

3. *Find* $\sqrt[3]{8}$

   ANSWER 2

4. *Find* $\sqrt[3]{27}$

   ANSWER 3

5. *Find* $\sqrt{100} + \sqrt{25}$

   ANSWER 15

6. *Does* $\sqrt{-4}$ *exist?*

   ANSWER No — an even root needs $a \geq 0$.

7. *Find* $\sqrt[3]{-8}$

   ANSWER -2

Medium · the standard question

7 PROBLEMS

1. *Simplify* $\sqrt{72}$

   ANSWER $6\sqrt{2}$

2. *Simplify* $\sqrt{50}$

   ANSWER $5\sqrt{2}$

3. *Simplify* $\sqrt{12} \cdot \sqrt{3}$

   ANSWER 6

4. *Simplify* $\dfrac{\sqrt{18}}{\sqrt{2}}$

   ANSWER 3

5. *Simplify* $\sqrt{45}$

   ANSWER $3\sqrt{5}$

6. *Write* $\dfrac{6}{\sqrt{2}}$ *without a root below*

   ANSWER $3\sqrt{2}$

7. *Simplify* $2\sqrt{3} + 5\sqrt{3}$

   ANSWER $7\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. *Simplify* $\sqrt{98} - \sqrt{50}$

   ANSWER $7\sqrt{2} - 5\sqrt{2} = 2\sqrt{2}$

2. *Simplify* $\sqrt[4]{16}$

   ANSWER 2

3. *Simplify* $(\sqrt{5})^2 + \sqrt{16}$

   ANSWER $5 + 4 = 9$

4. *Write* $\dfrac{10}{\sqrt{5}}$ *without a root below*

   ANSWER $2\sqrt{5}$

5. *Simplify* $\sqrt{2} \cdot \sqrt{8} - 1$

   ANSWER $\sqrt{16} - 1 = 3$

6. *Show that* $\sqrt{9 + 16} \neq \sqrt{9} + \sqrt{16}$

   ANSWER $5 \neq 7$

7. *Simplify* $\sqrt{a^2}$ *for* $a \geq 0$, *and for* $a < 0$

   ANSWER a when $a \geq 0$; $-a$ when $a < 0$.

07 Homework

Homework — 5 problems

Algebra 8, §8, pp. 39–41.

1. *Find* $\sqrt{144}$, $\sqrt[3]{64}$ *and* $\sqrt[3]{-125}$
2. *Simplify* $\sqrt{32}$

3. *Simplify* $\sqrt{20} \cdot \sqrt{5}$

4. *Write* $\frac{8}{\sqrt{2}}$ *without a root in the denominator*

5. *Simplify* $3\sqrt{7} - \sqrt{7}$